# MATLAB Modelling Record: AK-07 PCB Axial-Flux Motor

Oct 21, 2026 · @Olwethu

## 1. Purpose and status

MATLAB was used for system-level motor dynamics and steady-state performance of the machine as a three-phase electromechanical system. It was not used for field calculation, which was Maxwell's job, or for thermal analysis, which used Icepak and a Python network.

The model is a lumped, FEM-informed model. Its torque/back-EMF function $k(\theta)$ and its torque constant come from Maxwell, and MATLAB adds:

- the circuit equations;
- the rotor's mechanical dynamics;
- an idealised drive.

It investigated:

- three-phase back-EMF waveforms and torque against rotor angle, with sinusoidal and six-step currents;
- start-up and load-step dynamics under an ideal current-controlled drive with a speed PI loop;
- voltage-driven behaviour of the star winding, including inductance sensitivity;
- torque-speed envelopes and copper-loss-limited efficiency at a 12 V supply;
- a Rev0 / Rev1-S / Rev1-D design comparison.

It did **not** model a real controller (no PWM, field-oriented control, dead time or current sensing), losses other than copper, temperature, or the hardware.

| Category | What it covers |
| --- | --- |
| **Completed and executed in MATLAB** | Models 1, 2 and 3 and the design comparison, run on a UCT lab PC: Rev0 on 7 Oct 2026; Rev0, Rev1-S and Rev1-D on 12 Oct 2026 (final run 18:24–18:28) |

| Category | What it covers |
|---|---|
| **Cross-checked outside MATLAB** | The same equations re-implemented in Python (7 Oct), and Models 1 and 3 re-run in GNU Octave (8 and 11 Oct). These checks were made by Claude, not in MATLAB |
| **Analytical post-processing of Maxwell results (not MATLAB)** | Operating points (rev. B–D), Test D speed sweep, PWM ripple, thermal current limits, parallel-connection current sharing. All in Python, after the MATLAB work |
| **Planned or proposed, never done** | A Simulink model; a PWM/FOC controller model; re-running the MATLAB models with the measured resistance (1.89 Ω) and the Maxwell inductance (34–51 µH); any comparison with bench measurements |

## 2. Implementation

Plain MATLAB scripts were written and executed. **No Simulink model (.slx or .mdl) was built**, and the package contains none. No toolboxes are used; Model 2 uses the base-MATLAB solver `ode15s`.

**MATLAB version.** The exact release was not recorded. The output PNGs carry only the tag "MATLAB, The MathWorks, Inc.". The code needs R2016b or later because it uses local functions in scripts. If the report needs the release, read it from the UCT lab PC (the `ver` command), because the figures were produced on lab PC EBE-RED-018.

| Version | Date | Machines | K used | Status |
|---|---|---|---|---|
| `rev0_matlab` | 7 Oct 2026, run 22:34 | Rev0 only | 18.02 mN·m/A (exponential distance law, 5 mm region padding) | Superseded |
| `MOTOR_MATLAB_MODEL` v1 / v2 | 8 Oct 2026 | Rev0, Rev1-S, Rev1-D | v2: field map 19.50 / 41.64 / 60.28 mN·m/A | Superseded |

| Version | Date | Machines | K used | Status |
|---|---|---|---|---|
| `MOTOR_MATLAB_MODEL` v3 | 11 Oct 2026 | All three | Switchable: detailed board-by-board or field map | Superseded by v3.1 |
| **`MOTOR_MATLAB_MODEL` v3.1 (final)** | Code 12 Oct; **run in MATLAB 12 Oct 2026, 18:24–18:28** | All three, both K sources | Detailed 18.54 / 39.50 / 57.35 (headline); field map 19.50 / 41.64 / 60.28 (upper bound) | **Final; quote this one** |

An earlier v3 run at 11:19 on 12 Oct (before the 5th/7th harmonics were added) is in `MATLAB_results.zip`; the 18:24 run in `results_detailed` and `results_fieldmap` supersedes it.

**Final scripts** (folder `MOTOR_MATLAB_MODEL`):

- `run_all.m`: runs everything twice, once per K source, into `results_detailed` and `results_fieldmap`.
- `rev0_params.m`, `motor_params.m`: parameters.
- `motor_kfun.m`: the back-EMF/torque function.
- `motor_model1_current.m`, `motor_model2_voltage.m`, `motor_model3_performance.m`: the three models.
- `compare_designs.m`: the design comparison.

Section 9 describes each file.

## 3. Governing equations implemented

The equations are written in Maxwell's terminal convention, with θ the mechanical rotor angle and x = A, B, C.

**Torque/back-EMF function** (`motor_kfun.m`). For each phase the model uses k~x~(θ) = dλ~x~/dθ, in N·m/A = Wb/rad. It is built from the Maxwell single-board waveform: the fundamental plus 3rd, 5th and 7th harmonics, scaled to the chosen machine torque constant K. Phases B and C are phase A rotated by 30° and 60° mechanical (−120° / +120° electrical), with the small magnitude and angle errors measured in Maxwell (`K_scale` 1 / 1.0031 / 1.0003, `psi_err` 0 / 0.20 / 0.40°).

$$k_x(\theta)=K_{s,x}\sum_{n=1,3,5,7}\left[a_n\cos(n\,\alpha_x)+b_n\sin(n\,\alpha_x)\right],\qquad \alpha_x=p(\theta+\phi_x)-\delta_x$$

**Back-EMF and torque** (all models):

$$e_x=\omega\,k_x(\theta),\qquad T_e=\sum_x i_x\,k_x(\theta)$$

**Mechanics** (Models 1 and 2):

$$J\frac{d\omega}{dt}=T_e-T_L-B\,\omega,\qquad \frac{d\theta}{dt}=\omega$$

**Model 1, imposed currents** (ideal current-controlled drive; inductance not needed). The currents follow the k fundamental: sinusoidal i~x~ = I~q~·cos(α~x~), or six-step 120° blocks. A speed PI loop sets I~q~:

- K~p~ = J·ω~bw~/K~t~ and K~i~ = K~p~·ω~bw~/2, with ω~bw~ = 20 rad/s;
- a current limit with anti-windup;
- fixed-step RK4 integration, Δt = 0.1 ms.

**Model 2, voltage-driven star with an isolated neutral** (solved with `ode15s`, because τ~e~ ≈ 18 µs against τ~m~ ≈ 0.3 s):

$$L\frac{di_x}{dt}=v_x-v_n-R\,i_x-e_x,\qquad v_n=\frac{\sum v-\sum e}{3}\;\;(i_A+i_B+i_C=0)$$

- **2A:** rotor-synchronous sinusoidal phase voltages of fixed amplitude V, aligned with the back-EMF, with no speed loop.
- **2B:** six-step from the DC supply. Two phases conduct in series at an average voltage d·V~dc~: 2L·di/dt = d·V~dc~ − 2R·i − ω(k~a~ − k~b~). Reverse current is blocked, a simplified freewheel.

**Model 3, steady state** (no time integration). The current is on the q-axis (i~d~ = 0). The required phase peak voltage and the supply voltage limit are:

$$V_{ph,pk}=\sqrt{(\omega K+R I)^2+(p\,\omega L I)^2},\qquad V_{ph,pk}\le V_{dc}/\sqrt{3}\;\text{(SVPWM, modulation index 1)}$$

The torque constant and losses are:

$$K_t=1.5K,\qquad P_{Cu}=1.5\,R\,I_{pk}^2,\qquad \eta_{Cu}=\frac{T\omega}{T\omega+P_{Cu}}$$

The efficiency η~Cu~ counts copper loss only. Model 3 also derives K~m~ = K~t~/√(1.5R), and the acceleration time t~acc~ = J·ω/(K~t~·I~lim~) used in the comparison.

**Variables:**

| Symbol | Meaning | Unit |
|---|---|---|
| θ, ω | Rotor mechanical angle, speed | rad, rad/s |
| p | Pole pairs (8) | |
| i~x~, v~x~, e~x~ | Phase current, voltage, back-EMF | A, V, V |
| R, L | Phase resistance and inductance | Ω, H |
| J, B | Inertia, viscous damping | kg·m², N·m·s/rad |
| T~e~, T~L~ | Electromagnetic and load torque | N·m |
| K | Torque constant per phase (fundamental of k) | N·m/A |
| I~q~, I~lim~ | Commanded current, current limit | A pk |

**Inputs:** the Maxwell-derived k(θ) and K, together with R, L, J, B, V~dc~, I~lim~, the load torque and the speed reference (Section 4).

**Outputs:** waveforms of e, i, T~e~, ω and θ; the steady-state torque, ripple, copper loss and efficiency; and the speed limits (Section 6).

**Assumptions written into the code:**

- an ideal (average) inverter: no PWM ripple, dead time or switching loss;
- ideal current control in Model 1;
- phase-to-phase mutual inductance neglected;
- B = 0;
- no eddy, iron, magnet or bearing loss;
- R constant at its 20 °C model value;
- no saturation, so the torque is linear in current;
- harmonic phase independent of distance.

## 4. Input parameters (final run, 12 Oct 2026)

The values below are from the final v3.1 run. The headline K source is "detailed"; the field-map values were run as an upper bound. Source tags follow the code:

- **[AEDT]** solved in Maxwell;
- **[Py]** computed from Maxwell data;
- **[Est]** estimate;

- **[Assump]** assumption.

| Parameter | Rev0 | Rev1-S (stators in series) | Source and status |
|---|---|---|---|
| Torque constant K (per phase, T = 1.5·K·Î) | 18.54 mN·m/A (field map: 19.50) | 39.50 mN·m/A (field map: 41.64) | [AEDT/Py] Board-by-board detailed FEM sum, 20 mm padding; Rev1-S = S1 18.54 + S2 20.96 |
| Back-EMF constant K~e~ (phase, peak) | 0.0185 V/(rad/s): 1.376 V rms per 1000 rpm | 0.0395 V/(rad/s): 2.932 V rms per 1000 rpm | Same number as K (SI units); [AEDT] |
| Waveform harmonics | H3/H1 7.12 %; H5 about 0.3 %, H7 about 0.1 % (upper estimate) | H3/H1 6.70 %; H5/H7 × 2 boards | [AEDT] Single-board sweep (REV0_REF_ROTOR_MS_SWEEP) and the 8-angle run REV0_H8_D100_P20 |
| Phase balance | K~B~/K~A~ 1.0031, K~C~/K~A~ 1.0003; angle errors 0.20°, 0.40° | Same | [AEDT] Three-phase check (REV0_CHECK_PHB/PHC) |
| Phase resistance R | 1.389 Ω (5 × 0.2777 Ω) | 2.777 Ω | [AEDT] Maxwell Gate 1, copper tracks only, 20 °C; **excludes joints and leads. Later measured 1.87–1.90 Ω per stator, never used in MATLAB** |
| Phase inductance L | 24.9 µH (range 18.7–31.1) | 49.8 µH (37.3–62.3) | [Est] 5·L~board~ + 2·ΣM with L~board~ = 2.07 µH (Maxwell Gate 2A) and M~adj~ = 1.04 µH (two-board energy); Rev1-S = 2 × Rev0. **Later simulated in Maxwell (Test C, 20 Oct): 34–51 µH Rev0, 74–113 µH Rev1-S** |

| Parameter | Rev0 | Rev1-S (stators in series) | Source and status |
|---|---|---|---|
| Inertia J | $1.13\times10^{-4}$ kg·m² (rotor 227 g) | $2.22\times10^{-4}$ kg·m² (368 g) | [Assump] Hand calculation: 86/19 × 2 mm steel discs, 16 or 32 magnets at r = 28 mm, 19 mm shaft 50 mm long (Rev1-S + 11.1 mm). No bearings, coupling or load |
| Damping B | 0 | 0 | [Assump] Ideal; mechanical loss not modelled |
| DC supply V~dc~ | 12 V | 12 V | [Assump] Bench supply; SVPWM limit V~dc~/√3 phase peak |
| Current limit I~lim~ | 2 A pk | 2 A pk | [Assump] Set before any thermal result. Later candidate test points: 0.84 A pk (Rev0), 0.65 A pk (Rev1-S) |
| Load torque | 20 mN·m (load step) | 20 mN·m | [Assump] Demonstration load |
| Speed | 1000 rpm reference (133.3 Hz); sweeps 250–4000 rpm | Same | [Assump] Design target |
| Model 2 voltage | 2.5 V phase peak (sine); six-step duty 0.30 of 12 V | Same | [Assump] Chosen to give about 1300 rpm no-load for Rev0 |
| Poles | 16 (8 pole pairs) | 2 × 16 magnets | [Spec] |

Rev1-S is built from the Rev0 single-board FEM. S1 is the Rev0 stack and S2 is added in series in the same sense; the S2 phase is confirmed equal to S1 by FEM (0.0°). With the "detailed" source the fundamental is scaled to exactly 39.50 mN·m/A, so the image coefficient c = 1.166 only affects the old "law" option.

**Winding connection:** star (Y) with an isolated neutral, and the Rev1-S stators in series per phase. The parallel connection and delta were **not** modelled in MATLAB.

# 5. Simulation cases actually run

Every case below was executed for Rev0, Rev1-S and Rev1-D, twice: once with the detailed K (headline) and once with the field-map K (upper bound). This was the final run on 12 Oct 2026. On 7 Oct, Rev0 alone was run with K = 18.02 mN·m/A. All cases use star connection, with the Rev1 stators in series.

| Case | Drive | Speed | Current / voltage | Load | Integration |
|---|---|---|---|---|---|
| **1A** Steady waveforms | Imposed currents: sinusoidal (aligned with the EMF) and six-step 120° blocks | Fixed 1000 rpm, two electrical periods | 1 A pk | None (torque against angle) | Algebraic, 2001 points |
| **1B** Start-up and load step | Imposed currents, speed PI (20 rad/s bandwidth), sine and six-step | 0 → 1000 rpm reference | Limit 2 A pk | 20 mN·m applied 1.0 s, removed 1.6 s | RK4, 0.1 ms, 2.2 s |
| **2A** Voltage-driven, inductance sensitivity | Rotor-synchronous sine voltage, no speed loop | Free (settles near 1270 rpm no-load for Rev0) | 2.5 V phase peak; L = low / nominal / high | 20 mN·m from 1.5 s | ode15s, 3 s |
| **2B** Six-step from the DC supply | Average duty 0.30 of 12 V (two phases in series) | Free | 3.6 V across the pair | 20 mN·m from 1.5 s | ode15s, 3 s |
| **3A** Current sweep | Sine and six-step | Steady state | 0.25, 0.5, 0.75, 1, 1.5, 2 A pk | – | Algebraic |
| **3B** Speed sweep | Sine, i~d~ = 0 | 250–4000 rpm in 250 rpm steps | Current set by load | 0, 10, 20, 40 mN·m | Algebraic |
| **3C/3D** Envelope and efficiency map | SVPWM limit V~dc~/√3, 12 V | 50–4500 rpm | Up to 2 A pk | Up to K~t~·I~lim~ | Algebraic grid 120 × 100 |
| **Comparison** | Same supply for all designs | 1000 rpm operating point | 2 A pk limit | 20 mN·m | Algebraic |

### Not run in MATLAB:

- the parallel or delta connection;
- the 0.84 / 0.65 A pk test points (those were computed in Python);
- the measured resistance;
- the Maxwell-simulated inductance;
- any temperature dependence;

- PWM switching;
- speeds outside 0–4500 rpm.

## 6. Outputs

The headline numbers below come from the final MATLAB run (12 Oct 2026, detailed K), copied from the `model*_summary.txt` and `compare_summary.txt` files. They use the 1.389 / 2.777 Ω model resistance, not the measured value (Section 7).

| Output (12 V supply, star) | Rev0 | Rev1-S | File |
|---|---|---|---|
| Back-EMF at 1000 rpm, phase / line | 1.377 / 2.384 V rms | 2.932 / 5.079 V rms | model1_summary.txt |
| Torque at 1 A pk: sine mean (ripple) | 27.84 mN·m (0.88 %) | 59.32 mN·m (0.85 %) | model1_summary.txt |
| Torque at 1 A pk: six-step mean (ripple) | 30.73 mN·m (14.8 %) | 65.46 mN·m (14.8 %) | model1_summary.txt |
| Start-up 0 → 90 % of 1000 rpm at 2 A pk | 0.197 s | 0.183 s | model1_summary.txt |
| At 20 mN·m, 1000 rpm: current / copper loss / copper-limited efficiency | 0.716 A pk / 1.061 W / 66.3 % | 0.336 A pk / 0.467 W / 81.7 % | model1_summary.txt |
| 2A (2.5 V sine): no-load / loaded speed | 1273 / 778 rpm, identical for L = 18.7, 24.9 and 31.1 µH | 601 / 379 rpm, identical for all L | model2_summary.txt |
| 2B six-step, duty 0.30: no-load / loaded speed, loaded ripple | 1105 / 564 rpm, 2.4 % | 526 / 279 rpm, 2.1 % | model2_summary.txt |
| Copper loss at 1 / 2 A pk | 2.08 / 8.33 W | 4.17 / 16.66 W | model3_summary.txt |
| Voltage-limited no-load speed / maximum speed at 20 mN·m (12 V) | 3568 / 3054 rpm | 1675 / 1448 rpm | model3_summary.txt |

| Output (12 V supply, star) | Rev0 | Rev1-S | File |
|---|---|---|---|
| Copper-limited efficiency at 20 mN·m: 500 / 1000 / 2000 rpm | 49.3 / 66.0 / 79.5 % | 68.8 / 81.5 % / beyond the voltage limit | model3_summary.txt |
| K~m~ (torque per √copper loss) | 0.0193 N·m/√W | 0.0290 N·m/√W | compare_summary.txt |
| Time 0 → 1000 rpm at 2 A pk (no load) | 0.213 s | 0.196 s | compare_summary.txt |

With the field-map K (upper bound), Rev0 gives 3393 rpm no-load and 68.3 %, and Rev1-S 1589 rpm and 83.1 %.

**Figures produced, per machine**:

| File | Content |
|---|---|
| `1A_back_emf_1000rpm.png` | Phase and line back-EMF |
| `1A_torque_vs_angle.png` | Sine against six-step torque |
| `1B_startup_load_step.png` | Speed, torque and currents during start-up and the load step |
| `1B_phase_currents_loaded.png` | Loaded phase currents |
| `2A_sine_voltage_startup.png` | Three inductance cases overlaid |
| `2A_loaded_currents_zoom.png` | Loaded currents, close-up |
| `2B_six_step_startup.png` | Six-step start-up |
| `3A_current_sweep.png` | Torque and copper loss against current |
| `3B_speed_sweep.png` | Back-EMF and required voltage against speed |
| `3D_efficiency_map.png` | Copper-limited efficiency map |

The comparison adds `C1_torque_speed_envelopes.png`, `C2_efficiency_20mNm.png` and `C3_relative_figures_of_merit.png`.

**Raw data (CSV):**

| File | Content |
|---|---|
| `1A_waveforms_1000rpm.csv` | k~A,B,C~, e~A,B,C~, e~AB~, sine and six-step torque against angle |
| `1B_startup_sine.csv`, `1B_startup_sixstep.csv` | t, θ, ω, T~e~, T~L~, i~A,B,C~, P~Cu~ |
| `2A_sine_voltage_Lnom.csv`, `2B_six_step.csv` | Model 2 time series |
| `3A_current_sweep.csv`, `3B_speed_sweep.csv`, `3C_envelope.csv` | Steady-state sweeps and envelope |
| `compare_table.csv` | 17 figures of merit per design |

## 7. Verification

The MATLAB outputs were checked three ways: against other code, against later Maxwell results, and against the one hardware measurement available. They were **not** compared with any bench test of the motor, because none exists yet. The torque constant K is an **input** taken from Maxwell, so agreement on K-proportional quantities checks the implementation, not the physics.

**Code checks (implementation):**

- **7 Oct:** Rev0 Models 1 and 2 in MATLAB reproduced an independent Python implementation of the same equations exactly (recorded in `rev0_matlab/README.txt`).
- **8 Oct and 11 Oct:** Models 1 and 3 and the comparison were re-run in GNU Octave and reproduced the MATLAB tables. Octave cannot run Model 2 (`ode15s`), so Model 2 was checked in the Python mirror.

**Against Maxwell** (later, independent Maxwell runs; differences are MATLAB minus Maxwell):

| Quantity | MATLAB (12 Oct) | Maxwell | Difference |
|---|---|---|---|
| Rev0 back-EMF at 1000 rpm, phase / line | 1.377 / 2.384 V rms | 1.372 / 2.377 V rms (rotating model, 17 Oct) | +0.4 % / +0.3 % |

| Quantity | MATLAB (12 Oct) | Maxwell | Difference |
|---|---|---|---|
| Rev1-S back-EMF at 1000 rpm, phase / line | 2.932 / 5.079 V rms | 2.925 / 5.066 V rms (rotating model, 19 Oct) | +0.2 % / +0.3 % |
| Rev0 torque at 1 A pk, sinusoidal | 27.84 mN·m | 27.78 mN·m (current sweep, Test C, 20 Oct) | +0.2 % |
| Rev1-S torque at 1 A pk | 59.32 mN·m | 59.15 mN·m (Test C) | +0.3 % |
| Linearity of torque with current (assumed in MATLAB) | Exact by construction | Within 0.01 % from 0.5 to 3 A pk (Test C) | Assumption confirmed |
| Sinusoidal torque ripple, Rev0 / Rev1-S | 0.88 % / 0.85 % | 1.26 % / 1.16 % (rotating model) | MATLAB lower by 0.3–0.4 percentage points |
| Phase H3/H1, Rev0 | 7.1 % (input) | 5.5 % (rotating model) | Input from the static fit, 1.6 points higher |
| Phase inductance, Rev0 / Rev1-S | 24.9 / 49.8 µH (estimate) | 34–51 / 74–113 µH (Test C) | MATLAB 27–51 % low; Model 2 shows the results do not depend on L in this range |

The small ripple difference arises because MATLAB uses the 5th/7th harmonics of board 1 only, scaled by the number of boards near a magnet face, while the rotating model resolves every board.

Six-step ripple (14.8 %), start-up times and the Model 2 speeds have **no** Maxwell counterpart. They were not independently verified.

**Against the analytical (Python) post-processing.** The voltage-limited no-load speeds agree once the different modulation index is allowed for:

- Rev0: 3568 rpm in MATLAB (m = 1) against 3390 rpm in Test D (m = 0.95); 3568 × 0.95 = 3390.
- Rev1-S: 1675 rpm against 1590 rpm.

**Against measurements.** The only measurement is the phase resistance: 1.87–1.90 Ω per stator against 1.389 Ω in MATLAB, which is 36 % higher (the joints and leads are not in the model). MATLAB therefore underestimates copper loss and overestimates efficiency.

Recomputed with the measured resistance at 60 °C (2.16 Ω per stator), the operating point at 20 mN·m and 1000 rpm becomes:

| | MATLAB | With measured resistance |
|---|---|---|
| Rev0 copper-limited efficiency | 66.0 % | 55.6 % |
| Rev1-S copper-limited efficiency | 81.5 % | 73.9 % |

This recalculation was done by hand, not in MATLAB. No back-EMF, torque or speed measurement exists yet.

## 8. Limitations and unverified parameters

| Item | How it was treated | Status and effect |
|---|---|---|
| Winding inductance | Estimated 24.9 µH (Rev0), 49.8 µH (Rev1-S) from single-board and two-board Maxwell values; phase-to-phase mutual neglected | **Estimate.** Maxwell later gave 34–51 µH and 74–113 µH. Negligible for the 133 Hz results (ωL about 0.04 Ω against R), but it governs PWM ripple, which MATLAB does not model |
| Resistance | Maxwell copper value 1.389 Ω, fixed at 20 °C | **Low by 36 %** against the measured 1.87–1.90 Ω. No temperature rise. MATLAB copper loss and efficiency are optimistic, and its speed limits under load slightly optimistic |
| Inertia | Hand calculation from the modelled discs, magnets and an assumed 50 mm shaft (+11.1 mm for Rev1-S) | **Estimate.** Excludes bearings, hub, coupling and load. Affects only the start-up and acceleration times |
| Mechanical loss | B = 0, no friction or windage | **Not modelled.** Shaft torque is overestimated by the bearing drag; 2–5 mN·m was assumed later, outside MATLAB |
| Inverter | Ideal average voltage source; no PWM ripple, dead time, conduction or switching loss; Model 1 ideal current control | **Not modelled.** Inverter efficiency is effectively 100 %; the six-step freewheel is simplified |

| Item | How it was treated | Status and effect |
|---|---|---|
| Losses | Copper only | Eddy loss (20–44 mW at 1000 rpm), magnet and iron loss excluded. Efficiency is a copper-loss upper bound |
| Thermal | None. Fixed R, no temperature, and a 2 A pk current limit set before any thermal result | **Not modelled in MATLAB.** Thermal work was done separately in Icepak and a Python network (see Section 9) |
| Torque constant | Input from Maxwell (detailed 18.54 / 39.50; field map 19.50 / 41.64) | Carries the ±5 % FEM method band; geometry is as modelled, not as built (air gaps, 80/25 mm backplates) |
| Harmonics | Static single-board fit; 5th/7th from board 1 × number of boards near a magnet face | Upper estimate. Ripple 0.3–0.4 points below the rotating-model value |
| Supply, current limit, loads | 12 V, 2 A pk, 20 mN·m, 2.5 V test voltage | **Assumptions** for demonstration, not test conditions. The bench test points were set later (0.84 / 0.65 A pk, 12 / 24 V) |
| Connection | Star, Rev1-S stators in series | Parallel and delta not simulated |

One minor inconsistency in the files: the final `model1_summary.txt` still prints "5th+ harmonics not resolved", although v3.1 includes the 5th and 7th. The text line was not updated; the calculation is correct.

## 9. Evidence

All files are unchanged copies from `MATLAB_Modelling.zip`, collected in `MATLAB_Evidence_AK07.zip` with a file index. There are **no .slx or .mdl files**, because no Simulink model was built.

**Final model (v3.1, run 12 Oct 2026), folder `01_code_final_v3.1`:**

| File | What it does |
|---|---|
| `run_all.m` | Runs Models 1–3 for Rev0, Rev1-S and Rev1-D, then the comparison, once per K source (detailed, then field map) |

| File | What it does |
|---|---|
| `rev0_params.m` | Rev0 parameters with source tags: single-board k(θ) Fourier coefficients, distance decay, B/C phase errors, R, the L estimate, the J hand calculation, V~dc~, I~lim~ |
| `motor_params.m` | Builds Rev0 / Rev1-S / Rev1-D; the "FEM INPUTS" block (K source, H3/H1, 5th/7th); R and L for two stators in series; J and mass |
| `motor_kfun.m` | k~x~(θ) for phases A, B and C |
| `motor_model1_current.m` | Model 1: imposed currents. Waveforms at 1000 rpm; start-up with speed PI and load step |
| `motor_model2_voltage.m` | Model 2: voltage-driven star with isolated neutral (ode15s). Sine drive with three inductances; six-step from the DC supply |
| `motor_model3_performance.m` | Model 3: current and speed sweeps, torque-speed envelope, copper-limited efficiency map |
| `compare_designs.m` | Design comparison table and figures |

**Results:** folder `02_results_final_12oct`, holding `results_detailed` (headline) and `results_fieldmap` (upper bound). Each has a summary text file, PNG figures and CSV data per design and model, plus `results_compare`.

**First version:** the Rev0-only model and its results (7 Oct 2026, K = 18.02 mN·m/A), in folders `03_code_rev0_initial_7oct` and `04_results_rev0_7oct`. It is superseded and kept only as evidence.

**Related work that is not MATLAB, and should not be described as MATLAB:**

- **Thermal:** Icepak steady-state models built and run by AEDT Python scripts (`REV0_ICEPAK_BUILD.py`, `REV1_ICEPAK_BUILD.py`, `run_*_ICEPAK_STEADY.py`).
- **Lumped thermal network:** a Python network calibrated to Icepak (`rev0_thermal_lumped.py`, in Thermal.zip).
- **Post-processing:** the operating points, speed sweep and current-sweep analysis (Python, in AK07_Final_Simulation_Results.zip).

**Suggested wording for the methodology:**

> A lumped electromechanical model of each machine was implemented in MATLAB (scripts, no Simulink), with the back-EMF/torque function k(θ) and torque constant taken from the Maxwell finite-element results. Three models were run for Rev0 and Rev1-S at a 12 V supply:
>
> - an imposed-current model (waveforms, start-up and load step under a speed loop);
> - a voltage-driven star-connected model solved with ode15s (including an inductance sensitivity study);
> - a steady-state performance model (torque-speed envelope and copper-loss-limited efficiency).
>
> Resistance, inductance and inertia were model values and estimates, not measurements. Inverter, mechanical, eddy and thermal effects were not included.